

Week 2 Status Report

CCP-CNS2026 — Problem 3: Campus Network

Course: Computer Networks (CNS2026)

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Week: 2 of 5

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GitHub: <https://github.com/mustafaasif050-sys/CCP-CNS2026-Problem3>

1. Summary

Week 2 focused on implementing the Campus Network topology in a network simulation environment and achieving basic connectivity across all three VLANs. All core networking objectives for this week have been successfully completed.

2. Work Completed This Week

Task	Status	Notes
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Network simulation tool setup	✓ Done	Cisco Packet Tracer 9.0 used
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Topology built with all devices	✓ Done	Router, Switch, 3 PC clients
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VLAN 10 (Students) configured	✓ Done	Port Fa0/1 on SW1
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VLAN 20 (Staff) configured	✓ Done	Port Fa0/2 on SW1
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VLAN 99 (Guests) configured	✓ Done	Port Fa0/3 on SW1
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Trunk port configured (SW1 → R1)	✓ Done	802.1Q trunk on Gi0/1
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Router-on-a-stick configured	✓ Done	Sub-interfaces on Gi0/0/0
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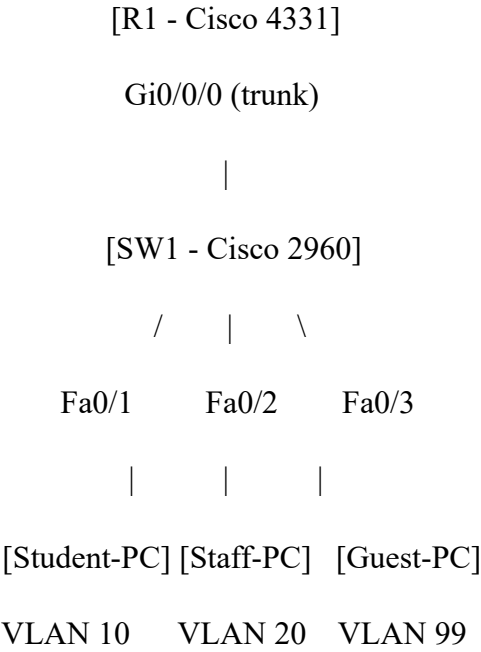
DHCP pools configured	✓ Done	All 3 VLANs getting IPs
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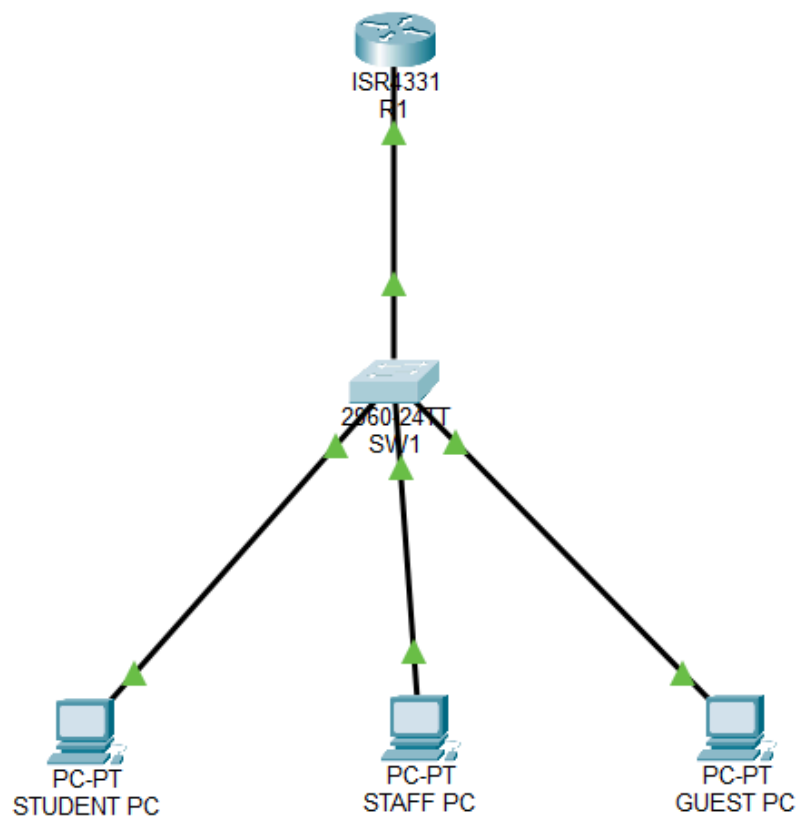
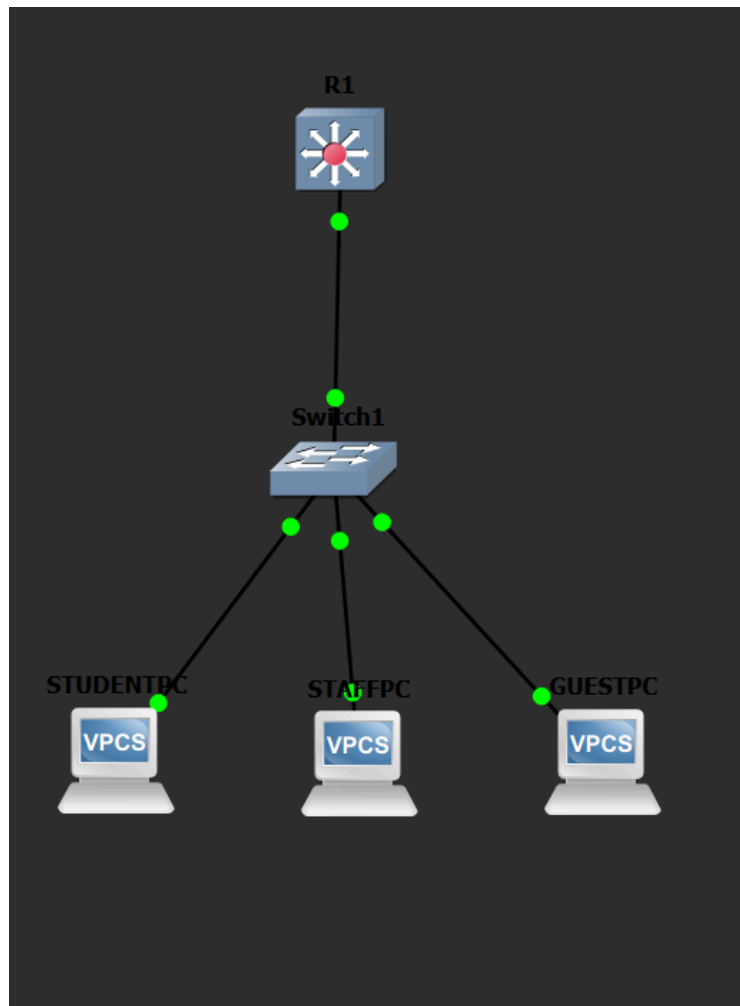
| **Inter-VLAN routing verified** |  **Done** | **0% packet loss** |

| **Guest isolation ACL applied** |  **Done** | **ACL 100 on Gi0/0/0.99** |

| **Running configs saved** |  **Done** | **R1-config.txt, SW1-config.txt** |

Topology Overview





IP Addressing — Verified

Device	Interface	IP Address	Status
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R1	Gi0/0/0.10	192.168.10.1/24	✓ Up
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R1	Gi0/0/0.20	192.168.20.1/24	✓ Up
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R1	Gi0/0/0.99	192.168.99.1/24	✓ Up
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Student-PC	Fa0	192.168.10.100/24 (DHCP)	✓ Up
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Staff-PC	Fa0	192.168.20.100/24 (DHCP)	✓ Up
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Guest-PC	Fa0	192.168.99.100/24 (DHCP)	✓ Up
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```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ip routing
R1(config)#interface gigabitethernet 0/0/0
R1(config-if)#no shutdown

R1(config-if)#exit
R1(config)#interface gigabitethernet 0/0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#exit
R1(config)#interface gigabitethernet 0/0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#exit
R1(config)#interface gigabitethernet 0/0/0.99
R1(config-subif)#encapsulation dot1Q 99
R1(config-subif)#ip address 192.168.99.1 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.10, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.20, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.99, changed state to up
```

```

Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Students
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Staff
Switch(config-vlan)#exit
Switch(config)#vlan 99
Switch(config-vlan)#name Guests
Switch(config-vlan)#exit
Switch(config)#interface fastethernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 99
Switch(config-if)#exit
Switch(config)#interface gigabitethernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#end
Switch#write memory
%SYS-5-CONFIG_I: Configured from console by console

```

5. Test Results

DHCP Test

Client	Expected Subnet	Result
Student-PC	192.168.10.x	✅ Got 192.168.10.100
Staff-PC	192.168.20.x	✅ Got 192.168.20.100
Guest-PC	192.168.99.x	✅ Got 192.168.99.100

IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.10.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	8.8.8.8

IP Configuration

DHCP

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

Static

192.168.20.100

255.255.255.0

192.168.20.1

8.8.8.8

IP Configuration

DHCP

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

Static

192.168.99.100

255.255.255.0

192.168.99.1

8.8.8.8

Ping Test Matrix

Source	Destination	Expected	Result
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Student-PC	192.168.10.1 (gateway)	✓	Success 0% loss
Student-PC	192.168.20.1 (Staff GW)	✓	Success 0% loss
Guest-PC	192.168.10.1 (Student GW)	✗	Blocked 100% loss
Guest-PC	192.168.20.1 (Staff GW)	✗	Blocked 100% loss

Guest Isolation — ACL Verification

ACL 100 applied inbound on Gi0/0/0.99:

- DENY: 192.168.99.0/24 → 192.168.10.0/24 ✓
- DENY: 192.168.99.0/24 → 192.168.20.0/24 ✓
- PERMIT: any → any (Internet access) ✓

```
R1(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.99
R1(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.99
R1(config)#ip dhcp excluded-address 192.168.99.1 192.168.99.99
R1(config)#ip dhcp pool STUDENTS
R1(dhcp-config)#network 192.168.10.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.10.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#exit
R1(config)#ip dhcp pool STAFF
R1(dhcp-config)#network 192.168.20.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.20.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#exit
R1(config)#ip dhcp pool GUESTS
R1(dhcp-config)#network 192.168.99.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.99.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#exit
R1(config)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

6. Rate Limiting Note

Rate limiting of the Guest VLAN to 5 Mbps was attempted using Cisco IOS QoS policy-map configuration. Cisco Packet Tracer 9.0 does not fully support the `police` command on sub-interfaces. On production hardware or GNS3 with a full IOS image, the following configuration would be applied:

```
class-map match-all GUEST_TRAFFIC
```


match any

policy-map RATE_LIMIT_GUEST

class GUEST_TRAFFIC

police rate 5000000 bps exceed-action drop

interface GigabitEthernet0/0/0.99

service-policy input RATE_LIMIT_GUEST

This configuration is documented as the intended production implementation.

7. Issues Encountered

| Issue | Resolution

| GNS3 VM incompatible with Windows 11 Hyper-V | Switched to Cisco Packet Tracer 9.0 |

| Named ACLs not supported in PT | Used numbered ACL 100 instead |

| Rate limiting not supported in PT | Documented intended config in report |

8. Next Week Plan (Week 3)

- Configure L2 security (DHCP snooping, port security)
- Implement captive portal simulation for Guest VLAN
- Configure and test rate limiting on real IOS
- Simulate attack and mitigation
- Capture PCAP files

9. GitHub Commits This Week

- Added topology .pkt file
- Added R1-config.txt and SW1-config.txt
- Added VLAN verification screenshots
- Added DHCP test screenshots
- Added ping matrix results